

AYUSH GAUR

AI/ML Engineer | Software Engineer | Data Scientist

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EDUCATION

University of Maryland, College Park, MD

Master of Science in Data Science

Relevant Courses: Advanced Machine Learning, Big Data Systems, Natural Language Processing, Deep Learning

Aug '24 - May '26

GPA: 3.89

Gautam Buddha University, Greater Noida, India

Bachelor of Technology in Computer Science Engineering

Jul '19 - Jul '23

GPA: 3.53

PROJECTS

Planera - Agentic Analytics Agent for Structured Data

Jan '26 - Feb '26

Python, FastAPI, DuckDB, SQLite, SQLAlchemy, LangGraph, React, TypeScript, Docker, JWT, Pytest, Vitest

- Built an inspectable chat-first analytics copilot for structured business data that lets users upload CSV/TSV/JSON files, ask questions in natural language, and inspect generated SQL, result previews, validation signals, and persisted conversation history in a protected workspace.
- Architected and productized the system end-to-end using a bounded LangGraph-based multi-agent workflow for schema-aware planning, SQL generation, DuckDB execution, and final synthesis, alongside JWT auth, user-scoped persistence, Dockerized deployment, and dual-provider support across OpenAI and Gemini.

Adaptive Diffusion Runtime Optimizer.

Jan '26 - Mar '26

Python, PyTorch, Hugging Face Diffusers, Stable Diffusion v1.5, CLIP, Ollama, Streamlit, Pytest

- Reduced Stable Diffusion v1.5 inference latency by 44.7% by engineering a prompt-adaptive inference pipeline that allocates denoising steps based on prompt complexity instead of fixed-step generation.
- Designed a reproducible evaluation framework using PyTorch, Hugging Face Diffusers, CLIP scoring, and latent-convergence early stopping to validate speed gains while preserving semantic alignment within a 0.0009 CLIP delta.

Contour - AI Sprint Planning & Jira Orchestration Copilot

Mar '26 - May '26

FastAPI, Next.js, React, TypeScript, OpenAI API, Pydantic, Zod, SQLite, Jira API

- Engineered a human-in-the-loop AI sprint planning copilot that turns sprint goals, messy backlog items, engineer skills, and capacity limits into Jira-ready plans with tasks, story points, owner recommendations, and delivery risk flags.
- Developed a production-safe Jira orchestration pipeline with typed validation contracts, constraint guardrails, dry-run previews, idempotency keys, and SQLite-backed sync state, validated across 50 synthetic sprint scenarios.

WORK EXPERIENCE

Data Scientist - Finsol Technologies Pvt. Ltd

Aug '23 - Jun '24

- Converted Excel-based financial records into structured, queryable datasets using Python, SQL, Pandas, NumPy, and Apache Spark, enabling CAs and risk analysts to analyze stock-market trends and support trader-facing reporting.
- Automated ETL workflows and Tableau dashboards for 35,000+ financial data points, improving data readiness for financial analytics, risk review, trend reporting, and ML experimentation.
- Built an internal RAG assistant using all-MiniLM-L6-v2, FAISS, and OpenAI GPT to replace manual document lookup for 400+ employees, enabling natural-language search over customer and personnel records with cited source retrieval.

Data Scientist Intern - XS INFOSOL Pvt. Ltd

Jul '22 - Sep '22

- Built backend services and SQL queries to support payroll and network bandwidth management workflows across 10 hotel chains, serving 50,000+ salaried employees and tracking 8,000+ room-level network connections.

SKILLS

- Languages:** Python, SQL, R, Java, C++, C, MATLAB
- Software Engineering:** Object-Oriented Design, System Design, REST APIs, Testing, CI/CD
- Backend / Cloud:** FastAPI, Docker, AWS, Linux, JWT, GitHub Actions
- Data / ML:** TensorFlow, scikit-learn, Hugging Face, LangGraph, LangChain, RAG, Generative AI, NLP, Computer Vision, CLIP, Stable Diffusion, Data Structures and Algorithms, Apache Spark, Airflow, Probability and Statistics
- Databases / Tools:** PostgreSQL, MySQL, SQLite, DuckDB, SQLAlchemy, MongoDB, Spark, Pandas, NumPy, Git, Pytest, Tableau, Power BI